

## Day 2 : Diagnosing Hardware FailuresTheory

**Task 1 : List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).**

| No. | Common Hardware Issue              | Typical Symptom                                                                                                 |
|-----|------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1.  | <b>Faulty RAM</b>                  | PC may fail to boot, show <b>no display</b> , restart repeatedly, or produce POST beep codes.                   |
| 2.  | <b>Power Supply (PSU) Failure</b>  | PC may <b>not turn ON at all</b> , or it may suddenly shut down/restart during operation.                       |
| 3.  | <b>Hard Disk/SSD Failure</b>       | System may become <b>very slow</b> , show boot errors, freeze frequently, or fail to load the operating system. |
| 4.  | <b>Overheating CPU</b>             | PC may become unusually hot, <b>slow down, freeze, or shut down automatically</b> under heavy load.             |
| 5.  | <b>Graphics Card (GPU) Problem</b> | Monitor may show <b>no display, artifacts, flickering, or driver-related crashes</b> while the PC is running.   |

**Task 2 : Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.**

| Step                                                      | Troubleshooting Action                                                                                                                     | What to Check                                                                        |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>1. Check the power outlet</b>                          | Plug another working device into the same socket.                                                                                          | Confirm that the wall socket is supplying power.                                     |
| <b>2. Inspect the charger</b>                             | Examine the adapter, cable, and connector for cuts, burns, loose connections, or physical damage.                                          | Look for visible damage that could prevent power delivery.                           |
| <b>3. Check the charging indicator</b>                    | Connect the charger and observe the laptop's charging LED.                                                                                 | If there is no LED indication, power may not be reaching the laptop.                 |
| <b>4. Check the charging port</b>                         | Inspect the laptop's DC charging port for dust, bent pins, looseness, or damage.                                                           | A damaged or loose port can prevent proper electrical contact.                       |
| <b>5. Test the charger with a multimeter</b>              | Set the multimeter to <b>DC voltage</b> and measure the charger's output at the connector.                                                 | Compare the measured voltage with the voltage printed on the adapter label.          |
| <b>6. Check the battery connection</b>                    | If the battery is removable, remove it and inspect its contacts; for an internal battery, inspect the connector only if safely accessible. | Check for loose connections, corrosion, or physical battery damage.                  |
| <b>7. Try charging without the battery (if removable)</b> | Remove the battery and connect the charger, then try powering on the laptop.                                                               | If the laptop works from the adapter without the battery, the battery may be faulty. |
| <b>8. Perform a basic power reset</b>                     | Disconnect the charger and battery (if removable), hold the power button for about <b>15–30 seconds</b> , then reconnect the power source. | This can clear residual electrical charge and reset basic power-state issues.        |

|                                                     |                                                                                                                                                    |                                                                                      |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>9. Test with a known-good compatible charger</b> | If available, try another charger with the <b>same required voltage, connector, and suitable current rating.</b>                                   | If the laptop charges with the other charger, the original charger is likely faulty. |
| <b>10. Identify the faulty section</b>              | Based on the results, determine whether the problem is with the <b>wall outlet, charger, charging port, battery, or internal charging circuit.</b> | This helps decide whether the component needs replacement or professional repair.    |

**Conclusion:** The troubleshooting should start with simple external checks and gradually move toward electrical testing with a multimeter. This helps identify the faulty component without using software diagnostics.

**Task 3 : Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).**

### **USB Charger Voltage Test Using a Multimeter**

#### **Tools Required**

- USB phone charger
- Digital multimeter
- USB cable

#### **Procedure**

1. Connect the USB phone charger to a working electrical power outlet.
2. Set the multimeter to **DC Voltage (V=)** mode.
3. If the multimeter has manual range selection, select the **20V DC** range.
4. Connect the **black probe** to the USB **GND (ground)** pin.
5. Connect the **red probe** to the USB **+5V (VBUS)** pin.
6. Observe the voltage displayed on the multimeter.
7. Record the measured voltage and compare it with the standard expected USB voltage range of **4.75V to 5.25V**.

#### **Observation**

| <b>Test</b>        | <b>Expected Voltage</b> | <b>Measured Voltage</b> | <b>Result</b>                       |
|--------------------|-------------------------|-------------------------|-------------------------------------|
| USB Charger Output | 4.75V – 5.25V           | <b>5.02V</b>            | <b>PASS – Within Expected Range</b> |

#### **Findings**

The USB phone charger produced a measured output voltage of **5.02V**. The expected voltage range for a standard USB 5V supply is approximately **4.75V to 5.25V**.

Since **5.02V falls within this range**, the charger output voltage is considered normal. Based on this voltage test, there is no indication of an abnormal output voltage from the charger.

#### **Conclusion**

The multimeter test showed that the USB charger is providing **5.02V**, which is within the acceptable USB voltage range. Therefore, the charger passed the voltage output test.

**Task 4 : A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?**

### **ATX Power Supply 12V Rail Voltage Test**

#### **Objective**

To test the **12V rail of an ATX power supply** using a multimeter and determine whether the power supply is providing an acceptable voltage.

#### **Tools Required**

- Digital multimeter
- ATX power supply
- ATX power connector

#### **Procedure**

1. **Switch off and unplug the PC** from the electrical outlet.
2. Disconnect the **ATX power connector** from the motherboard.
3. Set the multimeter to **DC Voltage (V=)** mode. If manual range selection is available, select the **20V DC range**.
4. Identify the wires on the ATX connector:
  - **Yellow wire = +12V**
  - **Black wire = Ground (GND)**
5. Place the **black multimeter probe** on a black ground wire.
6. Place the **red multimeter probe** on the yellow +12V wire.
7. Read and record the voltage displayed on the multimeter.
8. Compare the measured value with the expected 12V rail voltage.

#### **Observation**

| <b>Test</b>   | <b>Expected Voltage</b> | <b>Measured Voltage</b> | <b>Result</b>                         |
|---------------|-------------------------|-------------------------|---------------------------------------|
| ATX +12V Rail | Approximately 12V       | <b>12.08V</b>           | <b>PASS – Within acceptable range</b> |

## Findings

The measured voltage of the ATX +12V rail was **12.08V**. This indicates that the power supply is producing a voltage close to the expected 12V output.

If the measured voltage were **below 11.5V**, it could indicate a **weak or faulty power supply (PSU)** or a problem with its 12V output rail. An unstable or insufficient 12V supply can cause hardware such as the **CPU, motherboard, or graphics card** to receive inadequate power, which may result in random shutdowns, restarts, or system instability.

## Conclusion

The tested PSU showed a **12.08V reading**, which is close to the expected 12V output. Therefore, the 12V rail appears to be supplying normal voltage under the test conditions.